

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VI

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PART – A : Thoery

A **Binary Tree** is a non-linear data structure in which each node has at most two children: a **left child** and a **right child**.

Tree Traversal is the process of visiting every node of a binary tree exactly once in a specific order. The three basic traversal methods are:

- **Pre-order Traversal (Root → Left → Right):** Visits the root node first, then the left subtree, and finally the right subtree.
- **In-order Traversal (Left → Root → Right):** Visits the left subtree first, then the root node, and finally the right subtree.
- **Post-order Traversal (Left → Right → Root):** Visits the left subtree first, then the right subtree, and finally the root node.

These traversal techniques are commonly used for searching, displaying, and processing the nodes of a binary tree. They are usually implemented using **recursion**.

PART – B : Question

Tree Traversal: Write a program to:

- Implement pre-order,
- in-order,
- Post-order traversal of a binary tree.

Code:

```
File Edit Format Run Options Window Help
class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def inorder(root):
    if root:
        inorder(root.left)
        print(str(root.val) + "->", end='')

def postorder(root):
    if root:
        postorder(root.left)
        postorder(root.right)
        print(str(root.val) + "->", end="")

def preorder(root):
    if root:
        preorder(root.left)
        preorder(root.right)
        print(str(root.val) + "->", end="")

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("Inorder traversal")
inorder(root)
print("\npostorder traversal")
postorder(root)
print("\npreorder traversal")
preorder(root)
```

Output:

```
=== RESTART: C:/Users/nidhi/Downloads/data science/prac3.py ==  
Inorder traversal  
4->2->1->  
postorder traversal  
4->5->2->3->1->  
preorder traversal  
4->5->2->3->1->  
Nidhi S011  
>|
```